Binary Classification Model Performance Comparison: Evaluating the Accuracy of TabTransformers and Graph Neural Networks in Financial Fraud Detection

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Background & Motivation

Research Question

How do attention-based and graph-based architectures perform in detecting financial fraud?

- TabTransformer asks whether fraud lives in how features combine; GNN asks whether fraud spreads through how transactions connect

Dataset

- Source: IEEE-CIS Fraud Detection dataset (Kaggle / Vesta Corporation) — two raw files merged on TransactionID
- 590,540 financial transactions over ~6 months
- 3.5% fraud rate — severe class imbalance
- 434 raw features → reduced to 70 (50 numeric + 20 categorical) via LightGBM importance ranking
- Same 70 features used by both models — ensures a fair, direct comparison
- Primary metric: PR-AUC — at 3.5% fraud, a model predicting "not fraud" on every transaction achieves 96.5% accuracy.
- PR-AUC measures the precision-recall tradeoff across all thresholds and is robust to class imbalance. Random baseline ≈ 0.035

Experimental Setup

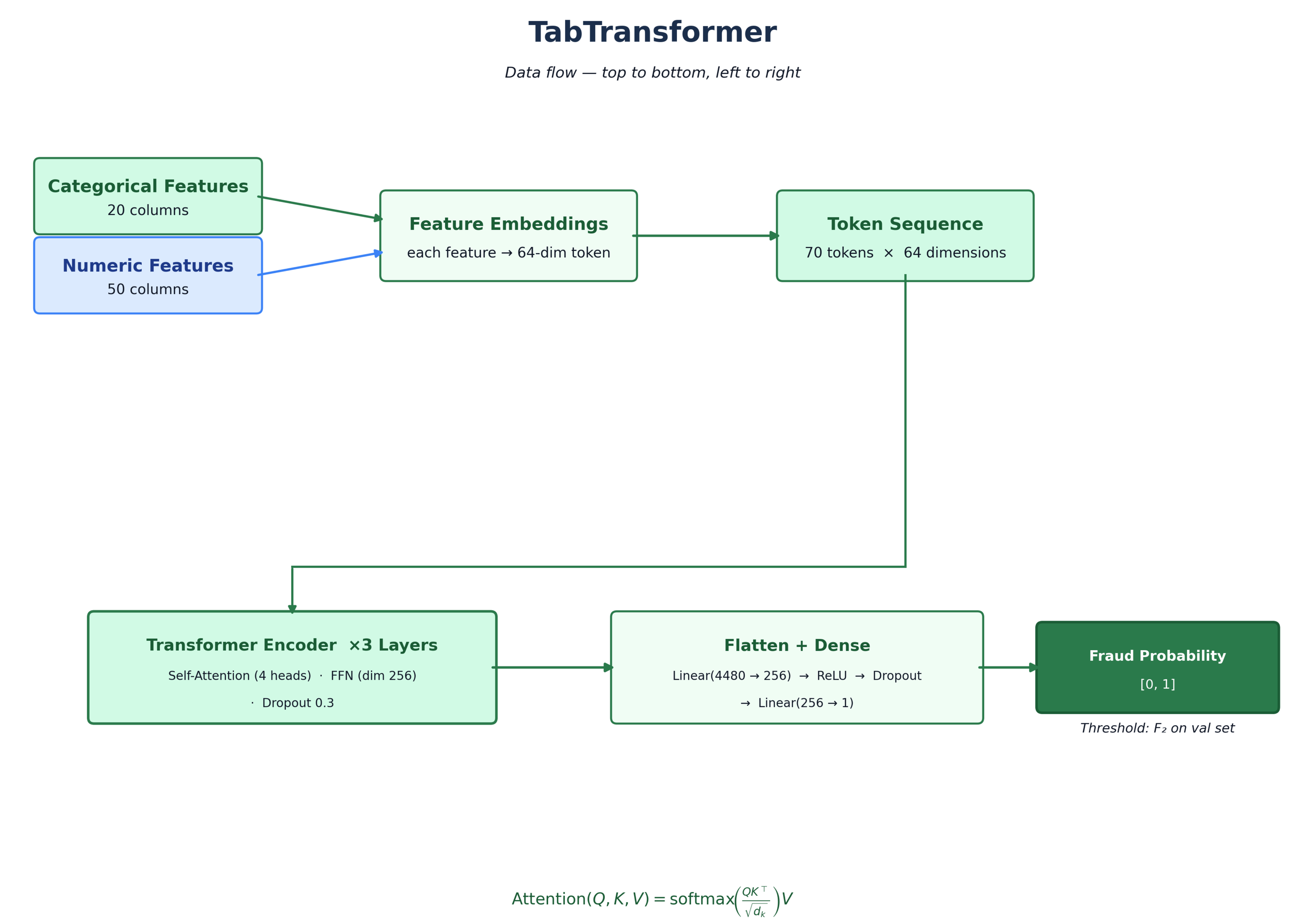
- Train / Validation / Test split: 70% / 15% / 15%, sorted by transaction time — models always train on older data and evaluate on newer, unseen transactions
- Decision threshold tuned on the validation set using F2 score, which weights recall twice as heavily as precision — in fraud detection, missing a fraud costs more than a false alarm
- Loss function: binary cross-entropy with ~27× class weighting to counteract the 3.5% fraud rate
- Label smoothing of 0.05 applied to both models to prevent overconfidence on noisy labels

Model Perspectives

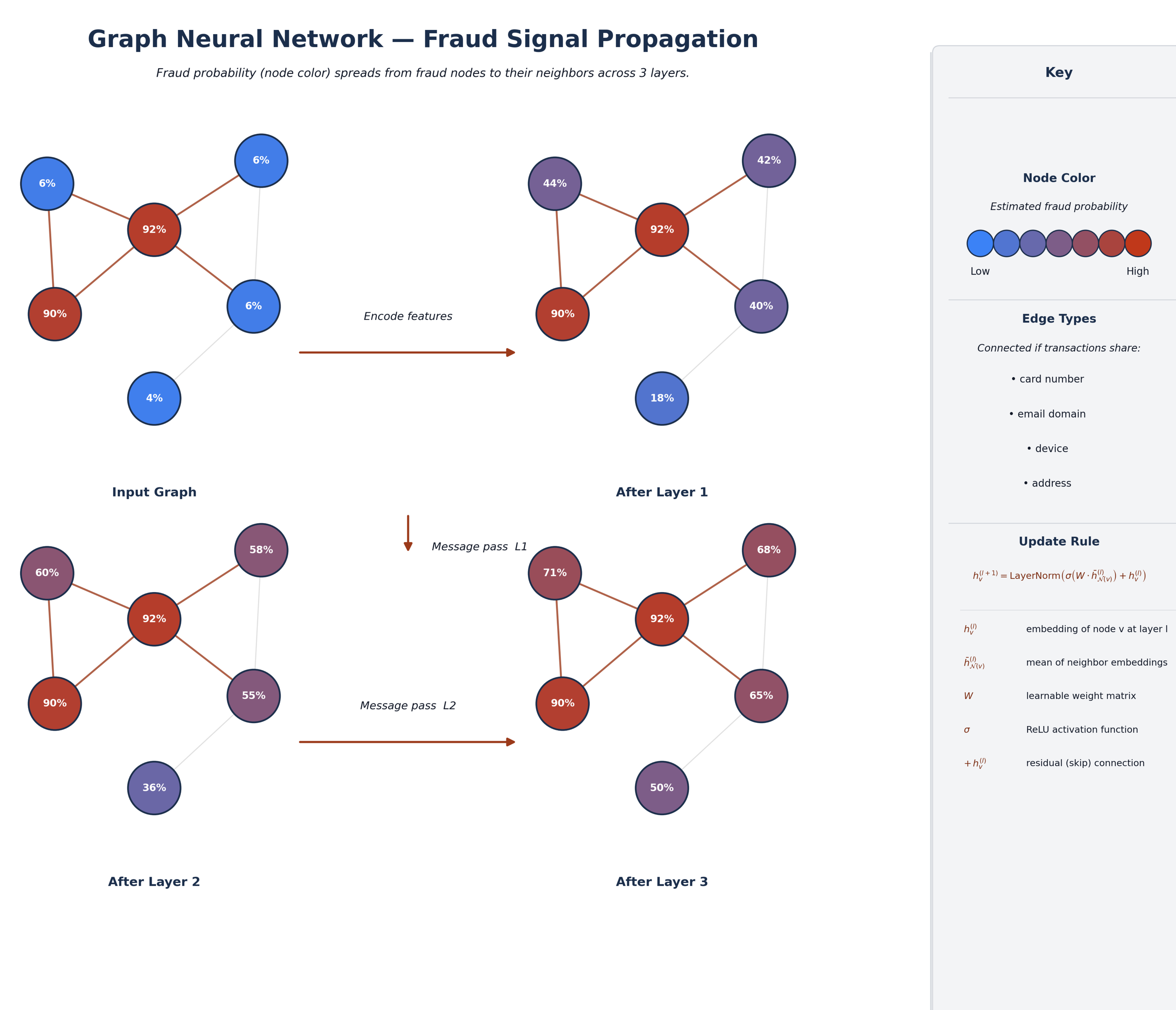
TabTransformer hypothesizes fraud lives in how features combine — attention captures joint patterns across cards, emails, and devices simultaneously. GNN hypothesizes fraud spreads through relationships — if a card was used fraudulently once, connected transactions deserve elevated suspicion. This dataset has both properties, making it an ideal testbed for a direct comparison.

Both models implemented entirely from scratch — no pretrained weights, no external model libraries.

Model Architectures

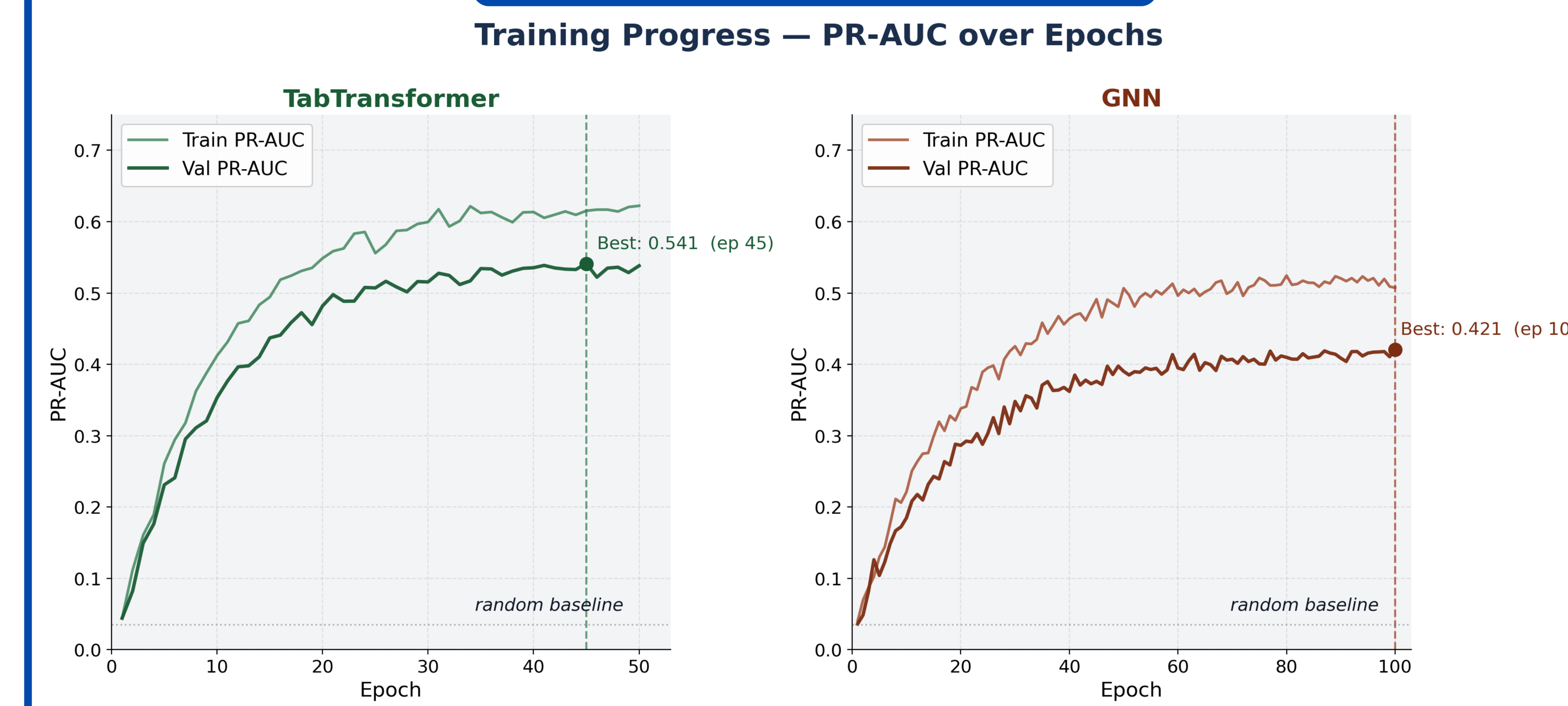


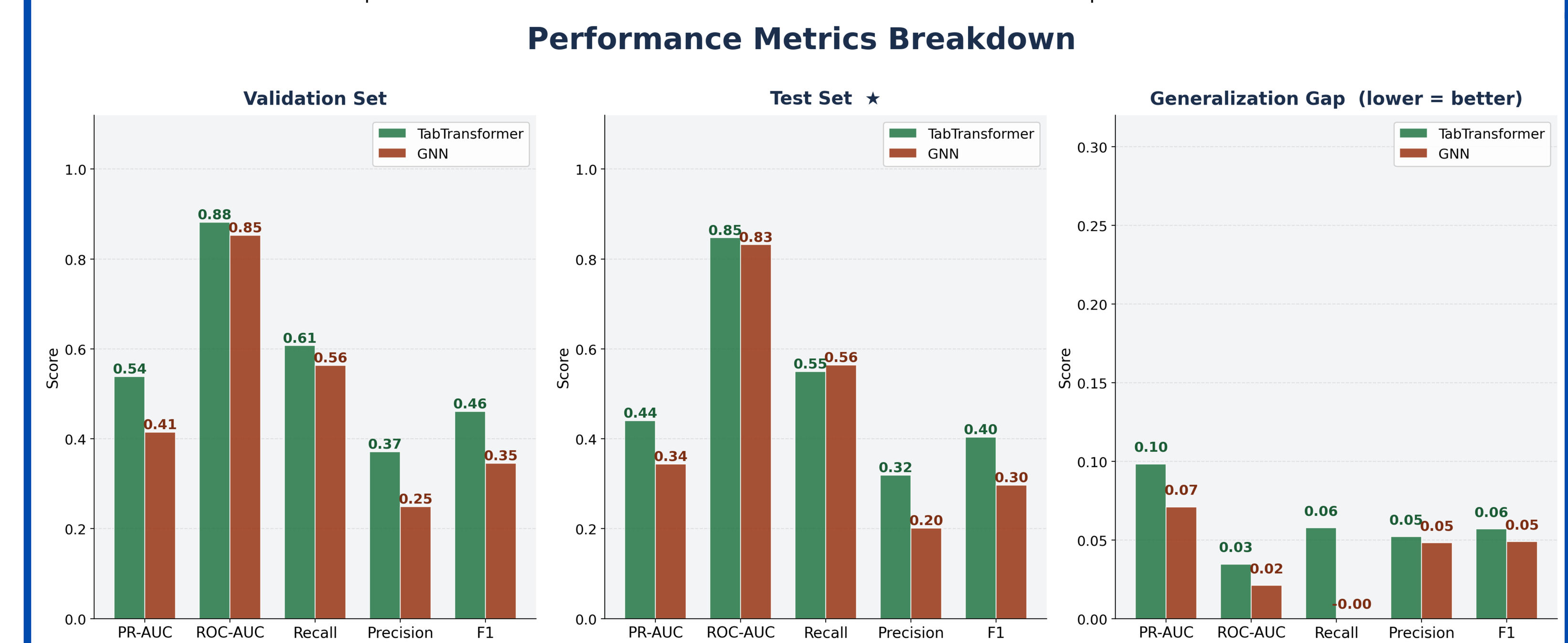
Encodes every feature — categorical and numeric — as its own 64-dim token, then applies 3 layers of self-attention (4 heads) to discover which feature combinations signal fraud without being told which matter. Output: single fraud probability per transaction.



Constructs a transaction graph — edges link transactions sharing a card, email, device, or address — then propagates fraud signals outward across 3 rounds of message passing. Residual connections ensure each node retains its own identity through aggregation. Output: fraud probability per node.

Performance & Conclusion





Results

Head-to-Head Comparison

TabTransformer vs. GNN · IEEE-CIS Fraud Detection · Test Set

Metric	TabTransformer	GNN	Winner
test PR-AUC ★	0.4402	0.3437	TabTransformer
test ROC-AUC	0.8471	0.8317	TabTransformer
test Recall	0.5495	0.5644	GNN
test Precision	0.3190	0.2012	TabTransformer
test F1	0.4036	0.2967	TabTransformer
Val→Test Gap ↓	0.098	0.071	GNN

★ Primary metric · ↓ lower is better · Highlighted cell = winning value

- Feature-level attention (TabTransformer) outperforms relational graph reasoning (GNN) on the primary metric — PR-AUC 0.44 vs. 0.34
- The GNN catches marginally more fraud (56% vs. 55% recall) but at the cost of significantly more false positives
- Both models perform 10–12× above the random baseline (PR-AUC ≈ 0.035) — meaningful signal exists in both approaches
- The two architectures are complementary — an ensemble combining feature-level attention with graph-level propagation is a natural direction for future work

Vaswani et al. (2017). "Attention Is All You Need." NeurIPS. · Dataset: IEEE-CIS Fraud Detection, Kaggle / Vesta Corporation.